

Design Specification

1. Project overview

`cpp-nn-visualizer` is a native C++17 raylib application for playing a legal human-vs-human chess game and inspecting neural-network activations for the current position. The chess core is independent of raylib and NN code so it can be unit-tested with `perft`, `FEN`, `SAN`, and `game-loop` tests.

2. High-level architecture

The executable links two static libraries:

- `cnv_core`: chess rules, game state, file I/O, tensor ops, and network inference.
- `cnv_viz`: raylib views and input handling. This layer owns no chess rules; it requests legal moves through `game::Game` and `chess::MoveGenerator`.

3. Module responsibilities

3.1 chess — legal-move core

The `chess` module owns board representation, FEN/SAN notation, move generation, make/unmake, draw/checkmate helpers, and Zobrist hashing. It is designed around bitboards for fast set operations plus a 64-square mailbox for simple UI and parser lookups.

3.2 game — game state and history

The `game` module owns the current `Position`, legal move application, status calculation, and the manually implemented linked-list move history. All mutation paths go through `game::Game` so the board, FEN snapshots, SAN text, undo, redo, and UI move list stay consistent.

3.3 nn — architecture-agnostic NN building blocks

The `nn` module contains shared tensor and activation infrastructure: `Tensor<T, Rank>`, `ActivationSnapshot`, `INetwork`, and low-level operations such as matrix multiplication, convolution, batch normalization, layer normalization, activation functions, and attention. CUDA kernels are optional; the CPU path remains the required runtime path.

3.4 nn/nnue — NNUE half-KP

The NNUE module implements feature encoding, accumulator refresh, model loading, and a forward pass that exposes accumulator values, clipped activations, active features, output weights, piece contributions, and a centipawn score.

3.5 nn/lc0_cnn — Leela CNN ResNet

The CNN module encodes the board into 112 input planes, runs a float32 LC0-style residual trunk, policy head, and WDL/value head, then stores intermediate feature maps in the shared activation snapshot for visualization.

3.6 nn/lc0_bt4 — Leela BT4 transformer

The BT4 module exposes a native C++ token-transformer visualization path. It uses the same board-plane encoder, projects the position to 64 square tokens, runs deterministic attention/FFN blocks, and produces policy/value summaries. This is not a protobuf-compatible LC0 BT4 weight loader; it validates the project's BT4 `.bin` metadata file and uses a C++ model path designed for explainable visualization.

3.7 viz — raylib UI

The viz module owns all raylib drawing and input handling: board rendering, piece sprites, controls, move list, FEN dialog, setup editor, activation views, promotion picker, theme, clocks, and asynchronous evaluation/search previews. It depends on `game`, `chess`, `nn`, and `io`, but those lower layers do not depend on raylib.

3.8 io — file I/O

The io module loads and saves configuration, FEN files, PGN text, and binary weight blobs. These are the project's main file-I/O paths and are used by both the user workflow and the grading requirement.

4. Class diagram

The main class diagram is provided as an image generated from `docs/class-diagram.dot`:

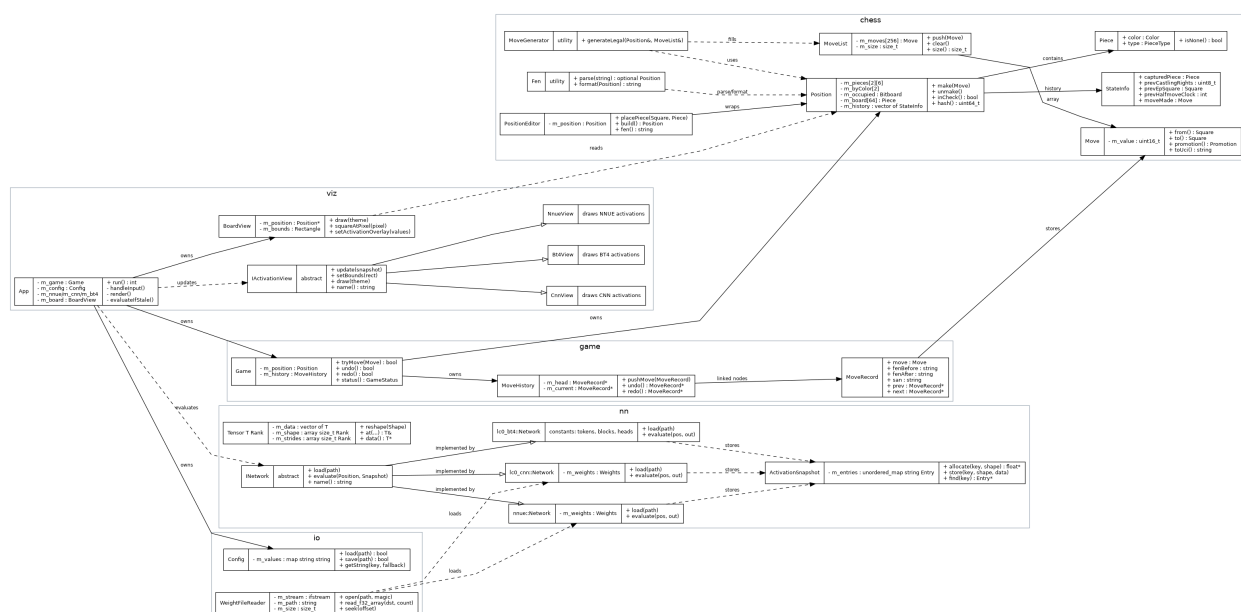


Figure 1: Core class diagram

4.1 Chess core class diagram

```
classDiagram
    class Color {
        <<enumeration>>
        White
        Black
    }

    class PieceType {
        <<enumeration>>
        None
        Pawn
        Knight
        Bishop
        Rook
    }
```

```

    Queen
    King
}

class Square {
    <<enumeration>>
    A1..H8
    None
}

class BitboardHelpers {
    <<namespace>>
    +popcount(Bitboard) int
    +lsb(Bitboard) int
    +popLsb(Bitboard&) int
    +setBit(Bitboard&, int) void
    +clearBit(Bitboard&, int) void
    +testBit(Bitboard, int) bool
}

class Piece {
    +Color color
    +PieceType type
    +isNone() bool
}

class Move {
    -uint16_t m_value
    +from() Square
    +to() Square
    +promotion() Promotion
    +toUci() string
    +parseUci(string) Move
}

class MoveList {
    -Move m_moves[256]
    -size_t m_size
    +push(Move) void
    +clear() void
    +size() size_t
    +swapErase(size_t) void
}

class StateInfo {
    +Piece capturedPiece
    +uint8_t prevCastlingRights
    +Square prevEpSquare
    +int prevHalfmoveClock
    +Move moveMade
}

class Position {
    -Bitboard m_pieces[2][6]

```

```

-Board m_byColor[2]
-Board m_occupied
-Piece m_board[64]
-Color m_sideToMove
-uint8_t m_castlingRights
-Square m_epSquare
-int m_halfmoveClock
-int m_fullmoveNumber
-vector<StateInfo> m_history
-vector<uint64_t> m_hashHistory
+pieceAt(Square) Piece
+make(Move) void
+unmake() void
+inCheck() bool
+hash() uint64_t
}

class Fen {
    <<utility>>
    +parse(string) optional<Position>
    +format(Position) string
}

class MoveGenerator {
    <<utility>>
    +generateLegal(Position&, MoveList&) void
}

class SlidingAttacks {
    <<namespace>>
    +bishopAttacks(int, Board) Board
    +rookAttacks(int, Board) Board
    +queenAttacks(int, Board) Board
    +knightAttacks(int) Board
    +kingAttacks(int) Board
    +pawnAttacks(int, Color) Board
}

class Perft {
    <<utility>>
    +perft(Position&, int) uint64_t
}

class San {
    <<utility>>
    +toSan(Position, Move) string
    +parse(Position, string) Move
}

class Zobrist {
    <<utility>>
    +computeZobristHash(Position) uint64_t
}

```

```

Piece --> Color
Piece --> PieceType
Move --> Square
MoveList o-- Move
StateInfo --> Piece
StateInfo --> Move
StateInfo --> Square
Position *-- StateInfo
Position --> Piece
Position --> Color
Position --> Square
Fen ..> Position
MoveGenerator ..> Position
MoveGenerator ..> MoveList
MoveGenerator ..> SlidingAttacks
Perft ..> MoveGenerator
San ..> MoveGenerator
Zobrist ..> Position

```

5. Data members

5.1 Chess core data member descriptions

Type	Member	Description
Piece	color	Side that owns the piece. Ignored when <code>type == PieceType::None</code> .
Piece	type	Piece kind, with <code>None</code> used for empty squares and parser failures.
Move	m_value	Packed 16-bit UCI move: bits 0-5 from-square, bits 6-11 to-square, bits 12-14 promotion code, 0xFFFF for no move. Castling and en-passant are inferred from the position.
MoveList	m_moves	Fixed array of 256 moves, large enough for the known legal-move upper bound and allocation-free in move generation.
MoveList	m_size	Number of valid entries currently stored in <code>m_moves</code> .
StateInfo	capturedPiece	Captured piece saved before <code>Position::make</code> ; <code>kNoPiece</code> if the move was not a capture.
StateInfo	prevCastlingRights	Four-bit castling-rights mask before the move.
StateInfo	prevEpSquare	En-passant target square before the move, or <code>Square::None</code> .
StateInfo	prevHalfmoveClock	Fifty-move counter before the move.
StateInfo	moveMade	Move being reversed by <code>Position::unmake</code> .

Type	Member	Description
Position	<code>m_pieces</code>	Twelve bitboards indexed by color and piece type, excluding <code>PieceType::None</code> . This is the primary fast representation for attack and occupancy queries.
Position	<code>m_byColor</code>	Two aggregate bitboards containing all white pieces and all black pieces.
Position	<code>m_occupied</code>	Union of both color occupancies.
Position	<code>m_board</code>	64-entry mailbox board used for $O(1)$ <code>pieceAt</code> lookups by UI, parsers, and make/unmake. It is redundant with the bitboards and must be kept in sync by <code>placePiece/removePiece</code> .
Position	<code>m_sideToMove</code>	Side whose legal moves are generated and whose king is checked by <code>inCheck</code> .
Position	<code>m_castlingRights</code>	Four-bit mask using <code>WhiteKing</code> , <code>WhiteQueen</code> , <code>BlackKing</code> , and <code>BlackQueen</code> .
Position	<code>m_epSquare</code>	Current en-passant target square, or <code>Square::None</code> when no en-passant capture is legal.
Position	<code>m_halfmoveClock</code>	Halfmove counter for the fifty-move rule. It resets after pawn moves and captures.
Position	<code>m_fullmoveNumber</code>	FEN fullmove number, incremented after each black move.
Position	<code>m_history</code>	Stack of <code>StateInfo</code> records that lets <code>unmake</code> restore irreversible state in $O(1)$.
Position	<code>m_hashHistory</code>	Sequence of Zobrist hashes since the current game/setup anchor. Repetition detection scans the reversible-move window.

Bitboard is a `uint64_t` value with A1 at bit 0 and H8 at bit 63. `Square`, `Color`, `PieceType`, `Fen`, `MoveGenerator`, `San`, `Perft`, `SlidingAttacks`, and `Zobrist` do not own persistent data beyond local temporaries and static lookup tables hidden inside their implementations.

5.2 Game and history data member descriptions

Type	Member	Description
<code>MoveRecord</code>	<code>move</code>	The move stored in this history node.
<code>MoveRecord</code>	<code>fenBefore</code>	Position before the move, used for reliable undo/jump reconstruction and debugging.

Type	Member	Description
MoveRecord	<code>fenAfter</code>	Position after the move, used for redo and move-list navigation.
MoveRecord	<code>san</code>	Human-readable move text displayed in the move history.
MoveRecord	<code>prev, next</code>	Manual doubly linked-list links for undo/redo traversal.
MoveHistory	<code>m_head</code>	First move in the current linear game line, or <code>nullptr</code> at game start.
MoveHistory	<code>m_current</code>	Most recently applied move; undo moves it toward <code>nullptr</code> , redo moves it toward the tail.
Game	<code>m_position</code>	Current legal chess position.
Game	<code>m_history</code>	Linked-list history synchronized with <code>m_position</code> .

5.3 Neural-network data member descriptions

Type	Member	Description
<code>Tensor<T, Rank></code>	<code>m_data</code>	Contiguous row-major data buffer.
<code>Tensor<T, Rank></code>	<code>m_shape</code>	Compile-time-rank array storing each dimension length.
<code>Tensor<T, Rank></code>	<code>m_strides</code>	Precomputed flat-index strides for fast <code>at(...)</code> access.
<code>ActivationSnapshot::Entry</code>	<code>shape</code>	Runtime shape of one activation tensor.
<code>ActivationSnapshot::Entry</code>	<code>data</code>	Flat float buffer for one activation tensor.
<code>ActivationSnapshot</code>	<code>m_entries</code>	Map from descriptive snapshot keys to activation entries.
<code>nnue::Network</code>	<code>m_weights</code>	Loaded or fallback NNUE weights, including feature weights, biases, output weights, and scaling.
<code>lc0_cnn::Network</code>	<code>m_weights</code>	Loaded LC0-style residual network weights, including trunk, policy, and value head data.
<code>lc0_bt4::Network</code>	static constants	Token count, model dimension, block count, attention head count, and policy size for the native transformer visualization.

5.4 I/O and visualization data member descriptions

Type	Member	Description
<code>Config</code>	<code>m_values</code>	Sorted map of flat INI-style key/value settings.

Type	Member	Description
WeightFileReader	m_stream	Open binary input stream for model files.
WeightFileReader	m_path	Source path used in diagnostics.
WeightFileReader	m_size	File size captured at open time for bounds/diagnostic checks.
BoardView	m_position	Non-owning pointer to the position currently being drawn.
BoardView	m_bounds	Rectangle occupied by the board in raylib screen coordinates.
BoardView	m_selection, m_legalTargets, m_lastMove	Visual overlays for interaction feedback.
BoardView	m_overlay	Per-square activation heatmap values.
App	m_config	Runtime configuration loaded from <code>config.ini</code> or an alternate config path.
App	m_game	Single owner of the playable game state.
App	m_board, m_controls, dialogs, and view pointers	Owned UI components that draw and handle user interaction.
App	m_nnue, m_cnn, m_bt4	Owned network evaluators for the architecture selector.
App	m_snapshot and m_evalCaches	Latest activation data plus per-architecture cached snapshots by Zobrist hash.
App	m_evalFuture, m_searchFuture	Background tasks so evaluation/search do not block the render loop.
App	m_editor and editor widgets	Temporary board setup state and validation UI.

6. Function members

6.1 Chess and game functions

Class / module	Important functions	Purpose
Position	clear, setStartpos, placePiece, removePiece, make, unmake	Maintain board state, apply legal moves, and restore previous positions.
Position	pieceAt, occupied, sideToMove, castlingRights, epSquare, hash	Read-only accessors used by move generation, UI, file I/O, and neural-network encoders.
Position	isSquareAttacked, inCheck, hasLegalMoves, isCheckmate, isStalemate, isInsufficientMaterial, isThreefoldRepetition	Rules and game-status helpers.
MoveGenerator	generateLegal(Position&, MoveList&)	Builds pseudo-legal moves and filters them by make/unmake and check validation.

Class / module	Important functions	Purpose
Fen	parse, format	Converts between standard six-field FEN strings and Position .
San	toSan, parse	Converts legal moves to/from SAN notation for the move list and tests.
Perft	perft(Position&, int)	Counts legal move trees for move-generator validation.
Game	reset, loadFen, tryMove, undo, redo, jumpToPly, status, statusText	Owns the playable game state and synchronizes position/history/UI behavior.
MoveHistory	pushMove, undo, redo, clear, size, plyCount	Manual doubly linked list used for undo/redo and move-list rendering.

6.2 Neural-network functions

Class / module	Important functions	Purpose
Tensor<T, Rank>	reshape, fill, at, data, shape, dim	Generic container used by NN code and tests.
ActivationSnapshot	allocate, store, has, find, data, shape	Network-agnostic storage for activations keyed by descriptive strings.
INetwork	load, evaluate, name	Polymorphic interface shared by all architecture implementations.
nnue::Network	load, evaluate, hiddenSize, isLoaded	Loads NNUE weights and publishes accumulator/output activations.
lc0_cnn::Network	load, evaluate, weights, isLoaded	Loads LC0-style CNN weights and publishes trunk, policy, and value tensors.
lc0_bt4::Network	load, evaluate, isLoaded	Runs the native BT4-style token visualization pipeline.
ops functions	matmul, conv2d, relu, mish, softmax, layer_norm, attention	Hand-written C++ NN operations used by the architecture implementations.

6.3 I/O and visualization functions

Class / module	Important functions	Purpose
Config	load, save, getString, getInt, getBool, setString, setInt, setBool	INI-style user configuration with typed accessors.
FenIo	loadPositionFromFenFile, savePositionToFenFile	Reads and writes FEN positions from text files.
WeightFileReader	open, read_u32, read_f32_array, seek, size	Little-endian binary model reader with schema validation handled by callers.

Class / module	Important functions	Purpose
App	run, handleInput, render, selectArchitecture, evaluateIfStale, saveCurrentFen, loadFenFromDialog	Top-level coordinator for raylib lifetime, UI state, game state, and asynchronous evaluation.
BoardView	draw, squareAtPixel, setSelection, setLegalDestinations, setActivationOverlay	Renders board, pieces, highlights, drag state, and activation overlays.
IActivationView	update, setBounds, draw, name	Polymorphic UI interface for NNUE/CNN/BT4 panels.

6.4 Neural-network evaluation flow

```
sequenceDiagram
    participant App
    participant Game
    participant Net as INetwork
    participant Snap as ActivationSnapshot
    participant View as IActivationView

    App->>Game: current Position
    App->>Net: evaluate(Position, Snapshot)
    Net->>Snap: store named tensors
    App->>View: update(Snapshot)
    View->>App: draw(theme)
```

NNUE, CNN, and BT4 share this path. **App** owns the concrete networks and activation views, but it talks to them through **INetwork** and **IActivationView**, which keeps the UI code independent from each network's specific tensor layout.

6.5 Teacher/TA review map

Rubric item	Where to inspect
Classes and encapsulation	src/chess/Position.*, src/game/Game.*, src/nn/Tensor.h, src/viz/App.*
Inheritance and polymorphism	src/nn/INetwork.h, src/viz/IActivationView.h, concrete NN/view classes
Templates	src/nn/Tensor.h
Arrays / fixed storage	src/chess/MoveList.*, bitboards, tensor buffers
Linked list	src/game/MoveHistory.*
File I/O	src/io/ConfigIo.*, src/io/FenIo.*, src/io/WeightFileReader.*, src/game/Pgn.*
Testing	tests/, scripts/test.sh, docs/test-cases.md
Demo evidence	docs/tutorial.gif, docs/demo.mp4

7. Main technical difficulties and solutions

7.1 Legal chess correctness

Chess move legality is difficult because castling, en-passant, promotion, checks, pins, and repetition all interact with state. The solution is to keep the fast bitboard representation and the simple mailbox representation

inside one `Position` class, then force all legal play through `MoveGenerator` and `Position::make/unmake`. Perft and rule-specific tests validate the result.

7.2 Undo/redo without corrupting game state

The game needs interactive undo/redo and move-list navigation. `MoveHistory` uses a manual doubly linked list of `MoveRecord` nodes, each storing move, SAN, FEN-before, and FEN-after. `Game` owns both `Position` and `MoveHistory`, so UI code cannot mutate the board without also keeping history consistent.

7.3 Neural-network activation volume

The application visualizes many thousands of activation values, not just a single score. `ActivationSnapshot` stores flat float buffers with explicit shapes. Each architecture publishes only the tensors that the UI needs, while views copy out compact summaries for drawing. This keeps the render loop responsive and avoids architecture-specific types leaking into the UI.

7.4 Translating prior chess work into C++

Some chess and NN ideas came from prior Java/TypeScript chess projects, but C++ required different ownership, header layout, memory handling, and build rules. The project uses RAII classes, explicit value types, standard containers, and CMake targets instead of directly copying runtime dependencies from the other projects.

7.5 Visual representation of dense models

Raw NN activations are too large to show literally. The UI therefore maps values into interpretable summaries: board overlays, heatmaps, bar charts, feature counts, policy/value summaries, token grids, and attention maps. The visual design was adjusted manually because generic AI-generated layout suggestions were often not usable for a raylib desktop tool.

8. File I/O justification

The project uses file I/O in several user-visible and implementation-critical ways:

File I/O path	Files	Reason
Configuration	<code>config.ini</code>	Window size, model paths, default FEN, clock, and theme settings can be changed without recompiling.
FEN save/load	<code>position.fen</code> and user-entered FEN files	Demonstrates text file input/output and supports reproducible board setup.
Model loading	<code>models/*.bin</code>	Binary file reading is required for NNUE/CNN/BT4-related model data.
PGN export support	<code>src/game/Pgn.*</code>	Converts played move history to standard chess notation for sharing games.
Test data	<code>tests/data/nnue_ref.jsonl</code>	Numerical tests compare C++ outputs against saved reference data.

The project therefore satisfies the mandatory file-I/O requirement through real application features rather than unrelated demonstration files.

9. Code quality and rationality

- Core rules, model logic, I/O, and rendering are separated into modules.
- Public members are limited; mutable state is mostly private and accessed through narrow methods.
- Inheritance is used only for real polymorphic boundaries (`INetwork`, `IActivationView`).
- Templates are used where they remove duplication (`Tensor<T, Rank>`).
- The hand-written linked list is isolated in `MoveHistory`, so the rest of the application can use a simple game-history interface.
- Global mutable state is avoided; long-lived state is owned by `App`, `Game`, or specific model/view classes.